Khadeeja Tapkirwala
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Education

University of Houston **January 2024 – December 2027**
Bachelor of Science in Computer Science, Data Science Concentration *Houston, Texas*
- Relevant Coursework: Object-Oriented Programming, Data Structures, Algorithms, Software Engineering, Database Systems, Computer Architecture, Discrete Mathematics

Experience

Structures & Artificial Intelligence Lab (SAIL), University of Houston **January 2026 – Present**
ML/AI Research Assistant *Houston, Texas*
- Develop object-oriented Python software, automation scripts, and ML pipelines for computer vision research.
- Build, test, and debug data processing workflows through experimentation, validation, and iterative development.
- Use Linux computing clusters and agentic AI tools including Claude Code and Codex to prototype and troubleshoot software.
- Document technical processes and collaborate with researchers to develop maintainable software solutions.

Modest Essentials, LLC **June 2024 – December 2025**
Operations Assistant / Sales Representative *Houston, Texas*
- Managed customer communications, inventory, fulfillment, and logistics while improving operational workflows.
- Resolved customer issues, coordinated orders, and supported process improvements across daily operations.

Stride, Inc. **September 2023 – March 2024**
Marketing Intern / Student Advisor Intern *Remote*
- Created digital content and collaborated with teams to support outreach and engagement initiatives.
- Conducted market research and analyzed engagement trends to improve communication strategies.

Leadership Experience

Cougarettes ACM-W, University of Houston **January 2026 – Present**
Vice President *Houston, Texas*
- Organize technical events and hackathons by coordinating sponsors, judges, volunteers, logistics, and student teams.
- Collaborate with students, faculty, and industry partners to deliver technical and professional development programs.

Project Experience

AI Resume Tailoring System **July 2026 – Present**
AI Software Project
- Develop an AI application using Python, NLP, transformer models, and LLM APIs to analyze job descriptions and resumes.
- Build an automated pipeline to rank experiences, generate tailored content, and produce ATS-friendly resumes.

MindFull Market — CodeRed Hackathon **October 2025**
AI Sustainability Web Application
- Built a web application using React, TypeScript, Vite, and the Google Gemini API to generate sustainability insights.
- Developed interactive data visualizations and user-facing features using Recharts.
- Used Git and GitHub collaboration workflows including pull requests, debugging, and iterative testing.
- Pivoted the system architecture after evaluating technical and platform constraints.

SFR Website — 1st Prize, CodeJam Hackathon **March 2025**
Web Development & UX/UI Project
- Designed and developed a user-focused website based on user needs, usability, and clear information flow.
- Combined UX/UI design with front-end web development to create an accessible and responsive user experience.
- Collaborated with teammates to iterate on the design and implementation based on feedback.

Technical Skills

Languages: Python, JavaScript, TypeScript, C++, R, HTML, CSS
Software Development: Object-Oriented Programming, Data Structures, Algorithms, SDLC, Debugging, Testing, Automation, Pytest
Web: React, Vite, Recharts, REST APIs
AI/ML: PyTorch, OpenCV, Computer Vision, NLP, Transformers, LLM APIs
Developer Tools: Git, GitHub, Linux, Claude Code, Codex, Google Gemini API